

Thermo-Mechanical Memory Controls Repeated Dike Failure During the Reykjanes Fires

0.1. Working concept

Target: A Nature / Nature Geoscience–style paper using the 2021–2026 Reykjanes volcanic-rifting sequence as a natural laboratory for a general physical mechanism.

Central hypothesis

> Each dike intrusion permanently modifies the thermo-mechanical state of the crust, so subsequent eruptions are controlled not only by magma recharge, but by the accumulated mechanical memory of previous intrusions.

The aim is **not** simply to build a thermo-mechanical model of Reykjanes. The aim is to identify and test a general mechanism for recurrent volcanic-rift failure.

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0.2. 1. The big idea

At Svartsengi and along the Sundhnúkur system, magma repeatedly accumulates at shallow crustal depths and is subsequently transferred into dikes. The 2021–2026 Reykjanes sequence gives us something unusually valuable: repeated magmatic and tectonic events occurring within essentially the same crustal system.

A simple conceptual model treats every event approximately as

$$\text{magma recharge} \rightarrow P_{\text{reservoir}} \rightarrow P > P_{\text{crit}} \rightarrow \text{dike}$$

In this picture, the crust is effectively reset between events and the critical failure pressure is approximately fixed.

Our alternative hypothesis is

magma recharge + tectonic stress + residual stress from previous intrusions + plastic deformation/damage + thermal weakening → failure

so that the critical state for the next dike depends on the complete history of the system.

In other words:

> **The crust remembers previous intrusions.**

This “memory” need not initially require an explicit damage-mechanics formulation. It can emerge from the state variables already available in a thermo-visco-elasto-plastic model:

- temperature (T),
- deviatoric stress (τ_{ij}),
- pressure (P),
- accumulated plastic strain (ϵ_{pl}),
- material distribution,
- viscosity ($\eta(T, P, \cdot, \epsilon)$),
- elastic stress history,
- and potentially evolving cohesion/friction.

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0.3. 2. Why Reykjanes is an exceptional natural laboratory

The Reykjanes Peninsula has experienced repeated unrest, diking and eruptions since 2021. Particularly important observations include:

- repeated shallow magma accumulation beneath Svartsengi;
- repeated rapid drainage into the Sundhnúkur dike system;

- a 15 km-scale dike during the November 2023 intrusion;
- repeated inflation–diking–deflation cycles;
- strong tectonic extension superimposed on magma-driven deformation;
- evidence that later intrusions can reuse previously established magma pathways;
- geochemical evidence for evolution from early crustally interacting magma toward magma transported through a more established plumbing system;
- GNSS and InSAR constraints on surface deformation;
- dense earthquake observations;
- seismic velocity changes associated with the intrusion sequence.

This means that different parts of the proposed physical model can potentially be tested independently.

The particularly interesting possibility is that the transition

difficult first intrusion → repeated pathway reuse

records the progressive mechanical conditioning of the crust.

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0.4. 3. Main scientific questions

0.4.1. Q1 – Does repeated intrusion lower the failure threshold?

Define a critical magma overpressure

$$\Delta P_{\text{crit}}^n$$

for intrusion number (n).

The first hypothesis to test is

$$\Delta P_{\text{crit}}^{n+1} < \Delta P_{\text{crit}}^n$$

The first intrusion creates or strongly modifies a pathway. Subsequent intrusions encounter a mechanically conditioned crust.

Eventually,

$$\Delta P_{\text{crit}} \rightarrow \Delta P_{\text{mature}}$$

corresponding to a mature plumbing system.

0.4.2. Q2 – Can thermal weakening eventually suppress brittle failure?

Repeated hot intrusions heat the surrounding crust. Initially this may promote localization and pathway reuse, but sufficiently hot crust may instead relax stress viscously.

This gives the possibility of non-monotonic evolution: ΔP_{crit} decreases and then increases.

There may therefore be a **thermo-mechanical window of maximum eruptibility**.

0.4.3. Q3 – Why does the system repeatedly select the Sundhnúkur pathway?

In 3-D, the preferred failure orientation should emerge from

$$\sigma_{ij}^{\text{total}} = \sigma_{ij}^{\text{tectonic}} + \sigma_{ij}^{\text{magma}} + \sigma_{ij}^{\text{previous intrusions}} + \sigma_{ij}^{\text{thermal}}$$

Rather than prescribing a dike trajectory, we want to determine whether the evolving stress/rheological architecture naturally favors the observed rift orientation.

0.4.4. Q4 – Is observed recharge volume a proxy for an evolving failure threshold?

Compare observed recharge between successive events with modelled

$$P_{\text{crit}}^n$$

or

$$V_{\text{crit}}^n$$

A constant-threshold model predicts approximately

$$V_{\text{crit}}^n \approx \text{constant}$$

The thermo-mechanical-memory model predicts

$$V_{\text{crit}}^n = f(T, \sigma_{ij}, \varepsilon_{\text{pl}}, \text{intrusion history}, \cdot V_{\text{m}})$$

This is a direct, falsifiable observational test.

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1. 4. Numerical strategy

1.1. Stage A — Minimal 2-D physics experiment

Start simple.

Preferred code: FEMTools.jl initially, because the unstructured FEM formulation makes reservoir/dike geometry and eventual fault structures particularly convenient.

Possible domain:

- width: 40–60 km;
- depth: 15–25 km;
- free surface;
- shallow reservoir at 4–5 km depth;
- extending crust;
- geothermal structure representative of Reykjanes.

1.1.1. Governing mechanical model

Use incompressible visco-elasto-plastic Stokes flow.

Conceptually,

$$\dot{\varepsilon}_{ij} = \dot{\varepsilon}_{ij}^{\text{el}} + \dot{\varepsilon}_{ij}^{\text{vis}} + \dot{\varepsilon}_{ij}^{\text{pl}}$$

Start with a Drucker–Prager-type yield criterion,

$$\tau_y = C + \mu P$$

Temperature-dependent viscous rheology:

$$\eta = \eta(T, P, \cdot \varepsilon)$$

Later extensions can include strain weakening:

$$C = C(\varepsilon_{\text{pl}}), \quad \mu = \mu(\varepsilon_{\text{pl}})$$

1.1.2. Tectonic boundary conditions

Impose regional extension approximately as

$$v_x(-\frac{L}{2}) = -\frac{V_{\text{ext}}}{2}$$

$$v_x\left(\frac{L}{2}\right) = +\frac{V_{\text{ext}}}{2}$$

Include gravity and a free surface.

1.1.3. Thermal model

Solve

$$\rho C_p \left(\frac{T'}{t'} \right) = \nabla \cdot (k \nabla T) + H$$

with heat advected if necessary once material motion becomes important.

Repeated intrusions provide transient thermal perturbations.

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2. 5. How to represent magma

This deserves explicit testing because we do not initially need full fracture mechanics.

Candidate approaches:

2.0.1. Approach A — Pressurized weak inclusion

Represent the magma reservoir as a low-viscosity/weak material region and increase its pressure or volume through recharge.

Advantages:

- straightforward;
- compatible with current thermo-mechanical machinery;
- excellent for initial parameter exploration.

2.0.2. Approach B — Volumetric magma injection

Add magma volume at a prescribed rate,

$$\cdot V_m$$

allowing pressure to emerge from the mechanical response.

This is physically more interesting because the model controls how much pressure accumulates versus how much deformation occurs.

2.0.3. Approach C — Explicit dike material insertion after failure

Let the model determine where failure/localization first occurs.

When a connected failure pathway satisfies a chosen criterion, convert that region into hot, weak magma/dike material.

Then evolve thermally and mechanically before the next recharge event.

This gives the desired cycle:

$$\text{recharge} \rightarrow \text{failure} \rightarrow \text{intrusion} \rightarrow \text{heating} \rightarrow \text{stress redistribution} \rightarrow \text{recharge}$$

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3. 6. Defining emergent failure

This is probably the most important methodological problem in the project.

We do **not** want to prescribe:

> “At 10 MPa overpressure, create a dike.”

Instead, failure should emerge from the stress field.

Possible criteria include:

3.0.1. Plastic connectivity

A dike-initiation event occurs when a connected yielded region extends from the magma reservoir toward the shallow crust.

3.0.2. Tensile criterion

Track the least compressive principal stress and identify regions where

$$\sigma_3 + P_f \geq T_0$$

where P_f is magma/fluid pressure and T_0 is tensile strength.

3.0.3. Combined shear/tensile criterion

Use Drucker–Prager or Mohr–Coulomb failure for shear localization together with a tensile cutoff.

This is probably the most physically defensible long-term direction.

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4. 7. Step-by-step work plan

4.1. Phase 0 — Literature and observational database

- Build chronology of the 2021–2026 Reykjanes sequence.
- Separate Fagradalsfjall and Svartsengi/Sundhnúkur episodes.
- Compile eruption and dike dates.
- Compile estimated dike dimensions.
- Compile reservoir depths.
- Compile inferred recharge volumes.
- Compile recharge rates.
- Compile GNSS displacement histories.
- Compile available InSAR products.
- Compile earthquake catalogs and hypocentre distributions.
- Compile published tectonic stress/orientation constraints.
- Compile crustal temperature/geothermal constraints.
- Compile seismic velocity-change observations.
- Compile petrological/geochemical evidence for pathway evolution.
- Create a single machine-readable event table.

Deliverable: `reykjanes_events.csv` + literature database.

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4.2. Phase 1 — Minimal mechanical model

Goal: demonstrate mechanically meaningful reservoir pressurization and failure without thermal evolution.

- Construct 2-D crustal domain.
- Add gravity.
- Add free surface.
- Implement regional extension.
- Initialize lithostatic pressure.
- Introduce reservoir at 4–5 km depth.

- Verify elastic/viscoelastic response to reservoir pressurization.
- Compare surface deformation with analytical/simple elastic solutions.
- Add Drucker–Prager plasticity.
- Add tensile cutoff if required.
- Determine first-failure pressure.
- Perform mesh-resolution tests.

Key output

$$P_{\text{crit}}^1$$

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4.3. Phase 2 — Repeated intrusion and mechanical memory

Now perform repeated cycles while initially keeping temperature fixed.

For each cycle:

1. recharge reservoir;
2. solve evolving stress;
3. identify failure;
4. emulate dike opening/intrusion;
5. relax the system;
6. recharge again.

Checklist:

- Implement accumulated plastic strain.
- Preserve stress history between events.
- Implement intrusion emplacement.
- Define robust automatic failure detection.
- Run 5–20 repeated cycles.
- Measure P_{crit}^n .
- Measure V_{crit}^n .
- Track pathway reuse.
- Track residual stress.
- Track accumulated plastic strain.
- Determine whether the system reaches a mature state.

Critical diagnostic

$$P_{\text{crit}}^n \quad \text{versus} \quad n$$

This tells us whether **purely mechanical memory** is enough to produce systematic evolution.

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4.4. Phase 3 — Add thermo-mechanical memory

Now switch on heat transport.

Each intrusion introduces hot magma into cold crust.

- Establish initial Reykjanes geothermal profile.
- Add realistic (k, ρ, C_p) .
- Insert hot dike material after intrusion.
- Evolve conductive cooling between events.
- Couple viscosity to temperature.

- Couple brittle/ductile transition to temperature.
- Repeat intrusion sequence.
- Compare with isothermal control runs.

Track

$$T(x, z, t)$$

$$\eta(x, z, t)$$

$$\tau_{II}(x, z, t)$$

$$\varepsilon_{pl}(x, z, t)$$

and

$$P_{crit}^n$$

Major question: does thermal evolution strengthen or weaken the tendency toward pathway reuse?

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4.5. Phase 4 — Determine the controlling physics

Run controlled numerical experiments where only one memory mechanism is retained.

4.5.1. Experiment 1 — No memory

Reset everything after each event.

4.5.2. Experiment 2 — Stress memory only

Retain τ_{ij} .

4.5.3. Experiment 3 — Plastic memory only

Retain ε_{pl} .

4.5.4. Experiment 4 — Thermal memory only

Retain T .

4.5.5. Experiment 5 — Full thermo-mechanical memory

Retain all state variables.

- Compare failure thresholds.
- Compare pathway geometry.
- Compare recurrence times.
- Quantify contribution of each mechanism.

This decomposition is extremely important because it lets us say **why** the crust remembers.

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4.6. Phase 5 — Parameter-space exploration

This is where JustRelax.jl becomes particularly attractive.

Explore:

- magma recharge rate ($\cdot V_m$);
- intrusion temperature;
- crustal geothermal gradient;
- tectonic extension rate;
- cohesion (C);

- friction coefficient (μ);
- viscous activation energy;
- reservoir depth;
- reservoir size;
- dike thickness;
- time between intrusions;
- magma/host-rock density contrast;
- initial stress state.

Potential dimensionless controls include

$$\Pi_T = \frac{t_{\text{recharge}}}{t_{\text{diffusion}}}$$

a Deborah-like number

$$D_e = \frac{t_{\text{Maxwell}}}{t_{\text{recharge}}}$$

and a magmatic/tectonic stress ratio

$$\Pi_\sigma = \frac{\Delta P_{\text{magma}}}{\sigma_{\text{tectonic}}}$$

- Identify appropriate nondimensional groups.
- Run broad 2-D parameter sweep.
- Classify failure regimes.
- Collapse results onto dimensionless scaling laws.